

Pilot and Trial Search Strings (Preliminary Search Iterations)

Before defining the final search strings used in the systematic literature review, several **pilot and trial search queries** were formulated and tested. These preliminary strings were used to (i) validate keyword coverage, (ii) assess recall and precision, and (iii) iteratively refine terminology aligned with the research questions. The pilot searches were executed on IEEE Xplore, Scopus, and ACM Digital Library.

Pilot String 1: Core Three-Area Intersection

This initial trial explicitly targeted all three areas using only canonical terminology.

("large language model" OR LLM OR ChatGPT OR GPT)
AND
("software architecture" OR "architectural design")
AND
("self-adaptive system" OR "self-adaptive software systems")

Observation

Returned very few results; many relevant studies used alternative terminology for self-adaptation or architectural activities.

Pilot String 2: Expanded Software Architecture Design Keywords

LLMs, Architectural design and reasoning terms were added to capture software architecture activities beyond high-level mentions.

("large language model" OR LLM OR "generative AI" OR ChatGPT OR GPT)
AND
("software architecture" OR "architecture design" OR
"architectural reasoning" OR "architecture evaluation")
AND
("self-adaptive system" OR "self-adaptive software systems" OR "SASS")

Observation

Improved architectural relevance, but several adaptive systems papers used different adaptation-related terminology.

Pilot String 4: Expanded Self-Adaptation and Autonomy Concepts

To reflect common terminology in SASS and CASA literature, all terms were generalized using wildcards.

("large language model" OR LLM OR "generative AI" OR GPT OR ChatGPT OR "Transformer")

AND

("software architecture" OR "architecture design" OR "architecture reasoning" OR "architecture evaluation" OR "architectural modeling")

AND

(adapt* OR self*)

Observation

This pilot achieved a better balance between recall and precision and became the structural basis for the final query.

Note

The pilot search strings were exploratory and used solely to refine keyword coverage and terminology. The final search string represents the stabilized version aligned with the research questions and was systematically adapted to each digital library.

Final Search String used for Review (2018-2025)

("large language model" OR "large language models" OR LLM OR LLMs OR

"generative AI" OR "generative Artificial Intelligence" OR "Gen AI" OR

"Gen Artificial Intelligence" OR "generative model" OR

"Transformer model" OR "Transformer models" OR

GPT OR "GPT-4" OR "GPT-3" OR ChatGPT OR BERT OR RoBERTa OR T5 OR PaLM OR LLaMA OR Claude)

AND

("software architecture" OR "architecture design" OR "architectural design" OR
"architecture modeling" OR "architectural modeling" OR "architecture evaluation" OR
"architecture analysis" OR "architectural reasoning" OR "architecture decision-making" OR
"design rationale" OR "decision support" OR "architecture framework" OR
"architecture pattern" OR "architecture tactic" OR "trade-off analysis" OR
"quality attribute analysis" OR "design decision")

AND (adapt* OR self* OR autonom*)

Note: This is the general query that we generated by extracting keywords from our Research Questions.

IEEE Xplore

(("Document Title": "large language model" OR "Document Title": "large language models" OR
"Document Title": "LLMs" OR "Document Title": "LLM" OR "Document Title": "generative AI" OR
"Document Title": "generative Artificial Intelligence" OR "Document Title": "Gen AI" OR
"Document Title": "Gen Artificial Intelligence" OR "Document Title": "generative model" OR
"Document Title": "Transformer models" OR "Document Title": "Transformer model" OR
"Document Title": "GPT" OR "Document Title": "GPT-4" OR "Document Title": "GPT-3" OR
"Document Title": "ChatGPT" OR "Document Title": "BERT" OR "Document Title": "RoBERTa"
OR "Document Title": "T5" OR "Document Title": "PaLM" OR "Document Title": "LLaMA" OR
"Document Title": "Claude" OR "Abstract": "large language model" OR "Abstract": "large language
models" OR "Abstract": "LLMs" OR "Abstract": "LLM" OR "Abstract": "generative AI" OR
"Abstract": "generative Artificial Intelligence" OR "Abstract": "Gen AI" OR "Abstract": "Gen
Artificial Intelligence" OR "Abstract": "generative model" OR "Abstract": "Transformer models"
OR "Abstract": "Transformer model" OR "Abstract": "GPT" OR "Abstract": "GPT-4" OR
"Abstract": "GPT-3" OR "Abstract": "ChatGPT" OR "Abstract": "BERT" OR
"Abstract": "RoBERTa" OR "Abstract": "T5" OR "Abstract": "PaLM" OR "Abstract": "LLaMA" OR
"Abstract": "Claude") AND ("Document Title": "software architecture" OR "Document
Title": "architecture design" OR "Document Title": "architectural design" OR "Document
Title": "architecture modeling" OR "Document Title": "architectural modeling" OR "Document
Title": "architecture evaluation" OR "Document Title": "architecture analysis" OR "Document
Title": "architectural reasoning" OR "Document Title": "architecture decision-making" OR
"Document Title": "design rationale" OR "Document Title": "decision support" OR "Document
Title": "architecture framework" OR "Document Title": "architecture pattern" OR "Document

Title:"architecture tactic" OR "Document Title":"trade-off analysis" OR "Document Title":"quality attribute analysis" OR "Document Title":"design decision" OR "Abstract":"software architecture" OR "Abstract":"architecture design" OR "Abstract":"architectural design" OR "Abstract":"architecture modeling" OR "Abstract":"architectural modeling" OR "Abstract":"architecture evaluation" OR "Abstract":"architecture analysis" OR "Abstract":"architectural reasoning" OR "Abstract":"architecture decision-making" OR "Abstract":"design rationale" OR "Abstract":"decision support" OR "Abstract":"architecture framework" OR "Abstract":"architecture pattern" OR "Abstract":"architecture tactic" OR "Abstract":"trade-off analysis" OR "Abstract":"quality attribute analysis" OR "Abstract":"design decision") AND ("Document Title":adapt* OR "Document Title":self* OR "Document Title":autonom* OR "Abstract":adapt* OR "Abstract":self* OR "Abstract":autonom*)) AND ("Publication Date":2018-01-01:2025-10-28)

Note: The studies retrieved using the IEEE Xplore search string can be found in a worksheet named “IEEE Xplore_LLMsSA_SASS” inside the file “Detailed_Analysis.xlsx” inside the replication package.

Direct link to the file:

https://github.com/NAZIA-SHAHZADI/LLMs4SAofSASS_ReplicationPackage/blob/main/Detailed_Analysis.xlsx

Scopus

(TITLE ("large language model" OR "large language models" OR "LLMs" OR LLM OR "generative AI" OR "generative Artificial Intelligence" OR "Gen AI" OR "Gen Artificial Intelligence" OR "generative model" OR "Transformer models" OR "Transformer model" OR GPT OR "GPT-4" OR "GPT-3" OR "ChatGPT" OR "BERT" OR "RoBERTa" OR "T5" OR "PaLM" OR "LLaMA" OR "Claude") OR ABS ("large language model" OR "large language models" OR "LLMs" OR LLM OR "generative AI" OR "generative Artificial Intelligence" OR "Gen AI" OR "Gen Artificial Intelligence" OR "generative model" OR "Transformer models" OR "Transformer model" OR GPT OR "GPT-4" OR "GPT-3" OR "ChatGPT" OR "BERT" OR "RoBERTa" OR "T5" OR "PaLM" OR "LLaMA" OR "Claude")) AND (TITLE ("software architecture" OR "architecture design" OR "architectural design" OR "architecture modeling" OR "architectural modeling" OR "architecture evaluation" OR "architecture analysis" OR "architectural reasoning" OR "architecture decision-making" OR "design rationale" OR "decision support" OR "architecture framework" OR "architecture pattern" OR "architecture tactic" OR "trade-off analysis" OR "quality attribute analysis" OR "design decision") OR ABS ("software architecture" OR "architecture design" OR "architectural design" OR "architecture modeling" OR "architectural modeling" OR "architecture evaluation" OR "architecture analysis" OR

"architectural reasoning" OR "architecture decision-making" OR "design rationale" OR "decision support" OR "architecture framework" OR "architecture pattern" OR "architecture tactic" OR "trade-off analysis" OR "quality attribute analysis" OR "design decision")) AND (TITLE (adapt* OR self* OR autonom*) OR ABS (adapt* OR self* OR autonom*))

Note: The studies retrieved using the IEEE Xplore search string can be found in a worksheet named “Scopus_LLMsSA_SASS” inside the file “Detailed_Analysis.xlsx” inside the replication package.

Direct link to the file:

https://github.com/NAZIA-SHAHZADI/LLMs4SAofSASS_ReplicationPackage/blob/main/Detailed_Analysis.xlsx

ACM/DL

(Title:"large language model" OR Title:"large language models" OR Title:"llms" OR Title:"llm" OR Title:"generative ai" OR Title:"generative artificial intelligence" OR Title:"gen ai" OR Title:"gen artificial intelligence" OR Title:"generative model" OR Title:"transformer models" OR Title:"transformer model" OR Title:"gpt" OR Title:"gpt-4" OR Title:"gpt-3" OR Title:"chatgpt" OR Title:"bert" OR Title:"roberta" OR Title:"t5" OR Title:"palm" OR Title:"llama" OR Title:"claude" OR Abstract:"large language model" OR Abstract:"large language models" OR Abstract:"llms" OR Abstract:"llm" OR Abstract:"generative ai" OR Abstract:"generative artificial intelligence" OR Abstract:"gen ai" OR Abstract:"gen artificial intelligence" OR Abstract:"generative model" OR Abstract:"transformer models" OR Abstract:"transformer model" OR Abstract:"gpt" OR Abstract:"gpt-4" OR Abstract:"gpt-3" OR Abstract:"chatgpt" OR Abstract:"bert" OR Abstract:"roberta" OR Abstract:"t5" OR Abstract:"palm" OR Abstract:"llama" OR Abstract:"claude")

AND

(Title:"software architecture" OR Title:"architecture design" OR Title:"architectural design" OR Title:"architecture modeling" OR Title:"architectural modeling" OR Title:"architecture evaluation" OR Title:"architecture analysis" OR Title:"architectural reasoning" OR Title:"architecture decision-making" OR Title:"design rationale" OR Title:"decision support" OR Title:"architecture framework" OR Title:"architecture pattern" OR Title:"architecture tactic" OR Title:"trade-off analysis" OR Title:"quality attribute analysis" OR Title:"design decision" OR Abstract:"software architecture" OR Abstract:"architecture design" OR Abstract:"architectural design" OR Abstract:"architecture modeling" OR Abstract:"architectural modeling" OR Abstract:"architecture evaluation" OR Abstract:"architecture analysis" OR Abstract:"architectural reasoning" OR Abstract:"architecture decision-making" OR Abstract:"design rationale" OR Abstract:"decision support" OR Abstract:"architecture framework" OR Abstract:"architecture

pattern" OR Abstract:"architecture tactic" OR Abstract:"trade-off analysis" OR Abstract:"quality attribute analysis" OR Abstract:"design decision")

AND

(Title:adapt* OR Title:self* OR Title:autonom* OR Abstract:adapt* OR Abstract:self* OR Abstract:autonom*)

Note: The studies retrieved using the IEEE Xplore search string can be found in a worksheet named “ACMDL_LLMsSA_SASS” inside the file “Detailed_Analysis.xlsx” inside the replication package.

Direct link to the file:

https://github.com/NAZIA-SHAHZADI/LLMs4SAofSASS_ReplicationPackage/blob/main/Detailed_Analysis.xlsx

Dataset Preparation

Overview

To prevent multiple counts of the same study, a deduplication process was applied to records retrieved from **IEEE Xplore**, **ACM Digital Library**, and **Scopus**, as the same article may be indexed in more than one database.

Data Collection

The initial dataset consisted of:

- **IEEE Xplore:** 61 records
- **ACM Digital Library:** 19 records
- **Scopus:** 196 records

This resulted in **276 retrieved records**, which were merged into a single dataset.

Duplicate Identification and Removal

Duplicates were identified through **exact matching of article titles**. Any study appearing in more than one database was considered a duplicate. All repeated occurrences and their source databases were documented in the worksheet “DuplicateArticles_LLMs_SA_SASS” in *Detailed_Analysis.xlsx*.

Link to the data:

https://github.com/NAZIA-SHAHZADI/LLMs4SAofSASS_ReplicationPackage/blob/main/Detailed_Analysis.xlsx

The duplicate analysis revealed:

- 55 article titles appearing multiple times
- 113 duplicate records in total

One instance of each duplicated article was retained, and the remaining **58 redundant records** were removed ($113 - 55 = 58$).

Deduplication Outcome

After duplicate removal:

- Initial records: 276
- Removed duplicates: 58
- **Unique studies retained: 218**

These 218 studies were used for screening.

Inclusion and Exclusion Criteria

Inclusion Criterion (IC)

A study was included only if it:

- Involved **Large Language Models (LLMs)**
- Addressed a **software architecture related problem**
- Related to **Self-Adaptive Software Systems (SASS)**
- Used LLMs as an **active component** in architectural design, reasoning, or decision-making

Exclusion Criteria (EC)

Studies were excluded if they were:

- **EC1:** Non-peer-reviewed (e.g., blogs, tutorials, theses)
- **EC2:** Not written in English

- **EC3:** Shorter than 5 pages

Study Selection Procedure

Step 1: Initial Retrieval

- Total retrieved studies: **276**

Step 2: Deduplication

- Duplicates removed: **58**
- Remaining unique studies: **218**

Step 3: Automatic Filtering

Database filters were applied for:

- Publication years: **2018–2025**
- Domain: **Computer Science & Engineering**
- Publication type: **Journals and conferences**

These filters restricted the dataset to relevant, peer-reviewed literature.

Step 4: Title and Abstract Screening

The remaining 218 studies were manually screened based on the inclusion criteria. A study was excluded if it:

- Used LLMs outside an architectural context
- Addressed software architecture without SASS
- Addressed SASS without LLM-based reasoning
- Focused on ML optimization without architectural relevance

Outcome

After deduplication, filtering, title and abstract screening, and full-text validation, **three (3) primary studies** were selected for the final analysis.